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| **Fiscal Year:** | FY 2024 | **Task Last Updated:** | FY 10/02/2025 |
| **PI Name:** | Tse, Stephen | | |
| **Project Title:** | Analysis of ISS Experimental Data from Response and Structure of Spherical Diffusion Flames | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Zip Code:** | 08901 | **Congressional District:** | 6 |
| **Comments:** | | | |
| **Project Type:** | Ground,Physical Sciences Informatics (PSI) | **Solicitation / Funding Source:** | 2023 NNH23ZDA001N-PSI E.8 Physical Sciences Informatics |
| **Start Date:** | 09/01/2025 | **End Date:** | 08/31/2027 |
| **No. of Post Docs:** | | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | 1 | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | | **Monitoring Center:** | NASA GRC |
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| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | 80NSSC25K0119 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| | The overall goal of the s-Flame experiment was to characterize and gain predictive capability of the structure and dynamics of diffusion flames in simple, well-defined flow fields such that the phenomena of interest can be studied without being unduly complicated and compromised by complex and sometimes non-quantifiable flow field effects. The basic flame configuration adopted for the investigation was the spherically symmetric diffusion flame generated by discharging a fuel gas from a porous spherical burner into a quiescent, oxidizing ambience in microgravity. The globally one-dimensional nature of the system facilitates data acquisition, data reduction, and computational simulation. The s Flame experiment used the Combustion Integrated Rack (CIR) and the Advanced Combustion via Microgravity Experiments (ACME) hardware in the Destiny module of the ISS.<br>The proposed project's objective is to process raw data from the s-Flame microgravity experimentation, conduct computational simulation, and perform theoretical analysis. Interpretation of all phenomena will emphasize the | | |

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| **Task Description:** | simultaneous and coupled influence of chemistry and transport (including radiation models). Specifically, the aims are: (i) Process imaging data, thin film pyrometry (TFP), and radiometer data to characterize the experimental structure and dynamics of the flames to compare with computational quasi-steady and transient simulations. Theoretical analysis will assess the quasi-steady nature of the flame dynamics. (ii) Run quasi-steady and transient simulations with different chemical kinetic mechanisms and different radiative loss models (e.g., adiabatic, optically thin, optically thick). (iii) Process imaging data, thin film pyrometry (TFP), and radiometer data to identify both kinetic and radiative extinction (i.e., at low and high system Damköhler (Da) numbers, respectively) to compare with transient simulations. Quasi-steady simulations will map the maximum temperature flame response with respect to Da number to identify extinction limits at the turning points to compare with transient experimental extinction. (iv) Process imaging data, thin film pyrometry (TFP), radiometer data, and PMT data to determine the existence onset, frequency, and amplitude of flame oscillations to compare with transient simulations to assess if the mode is that of the pulsating instability that has been theoretically predicted to occur in such flames with fuel/diluent mixtures that are above a critical Lewis number. Transient simulations compared to quasi-steady mapping of flame response with respect to Da number will identify the regions of pulsating or other instability near the extinction limits (turning points). (v) Perform ignition simulation and analysis to understand better the diluent effect on experimental ignition (e.g., where helium cases needed significantly higher ignition energy and power).<br><br>The main products of the proposed work will be (i) insights from fundamental characterization of transient dynamics and responses of such flames, (ii) categorization of processes that are either quasi-steady (e.g., expanding flame front motion toward steady state) or fully transient (e.g., pulsating instabilities prior to extinction), (iii) quantitative and qualitative mapping of extinction boundaries and unstable regimes, (iv) clarification of the role of diluents in various processes, (v) and additional understanding of ignition in nonpremixed flames. With diffusion flame burning as the basis in many combustion engines and devices, understanding the underlying dynamics and chemistry of diffusion flames is important in improving performance in terms of power, efficiency, pollutant emissions, and reliability. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | |
| **Task Progress:** | New project for FY24 |
| **Bibliography Type:** | Description: (Last Updated: ) |